In the Loop with Complex Analysis

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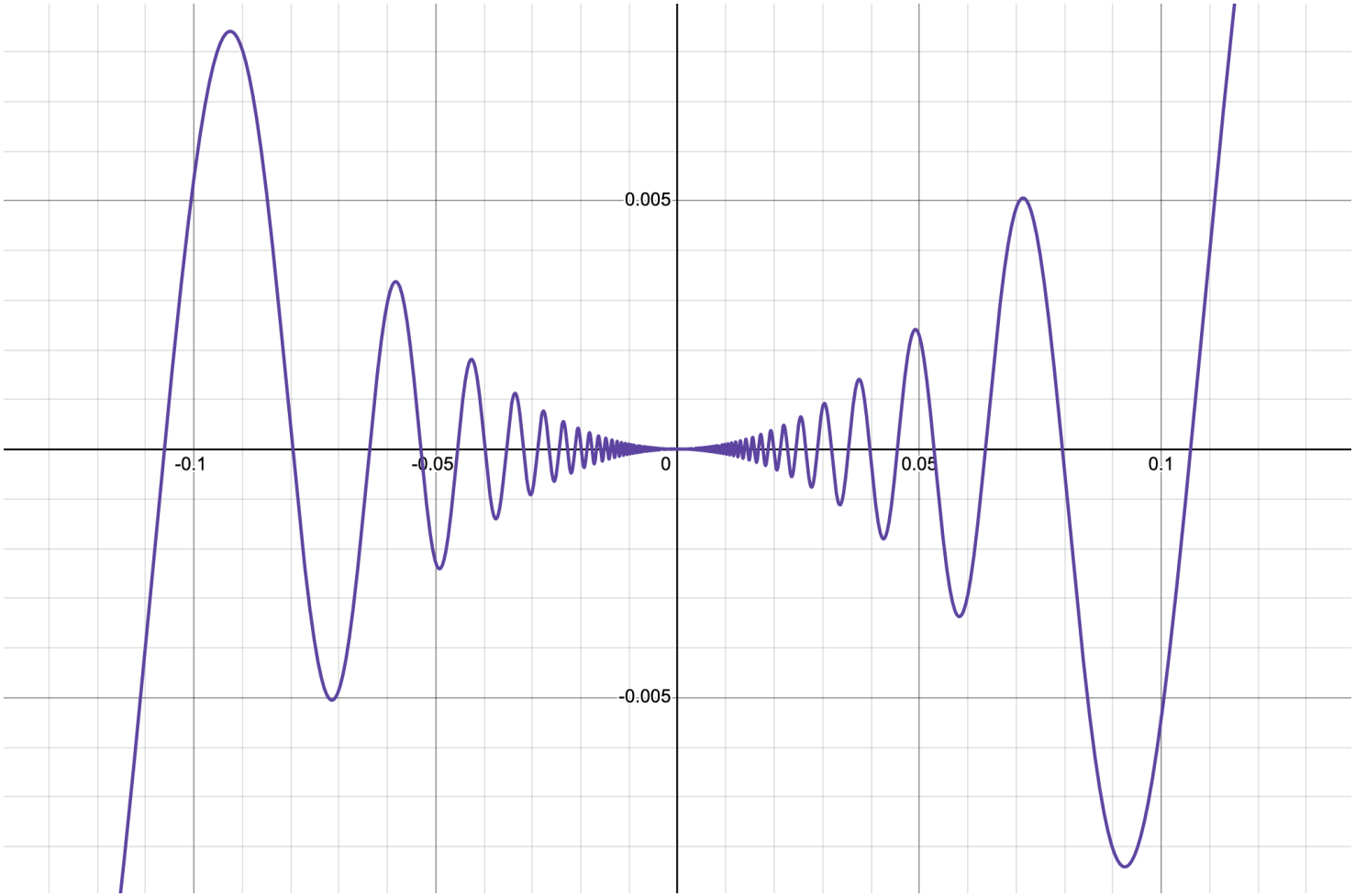
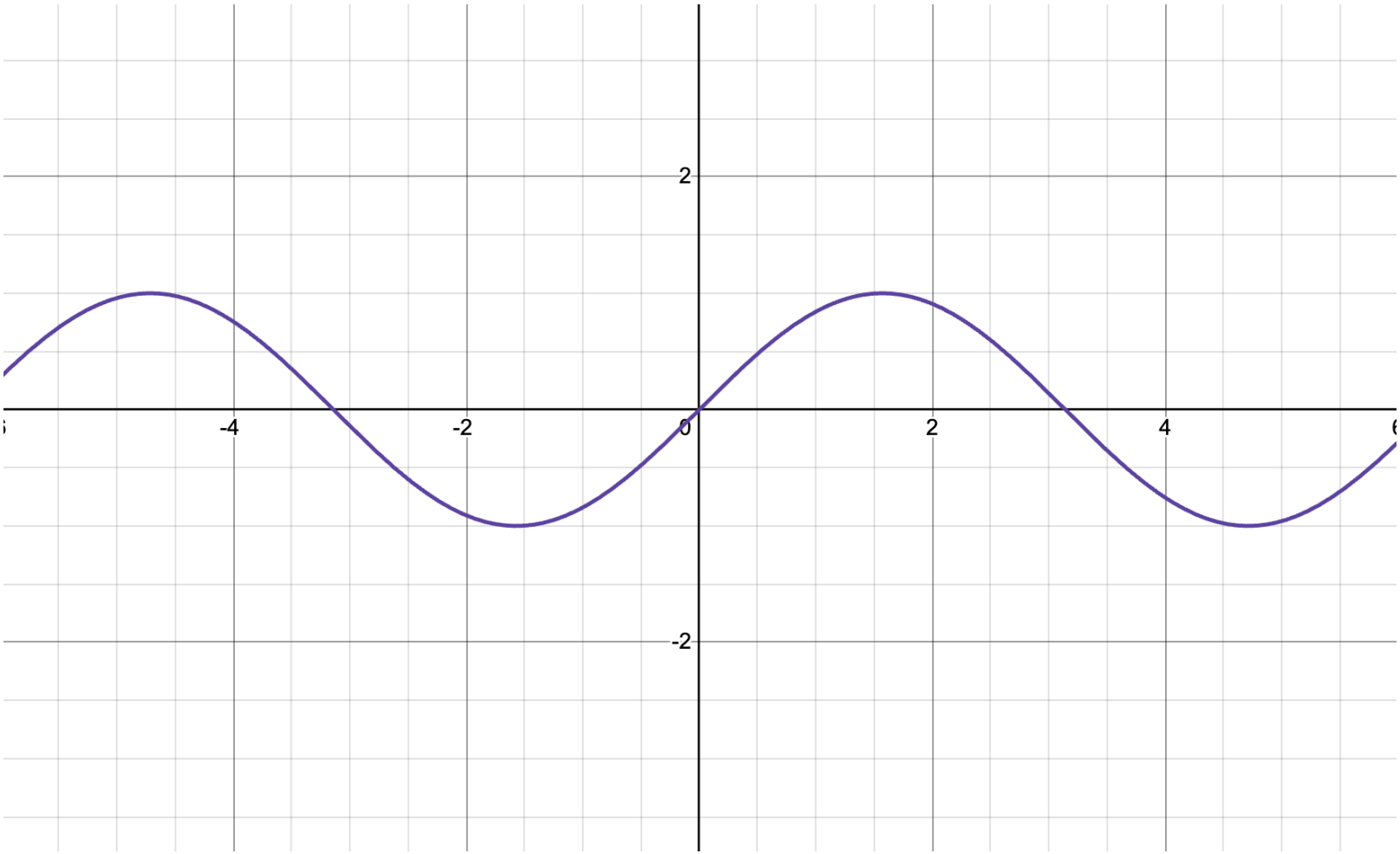


$x^2 \sin\left(\frac{1}{x}\right)$ is differentiable at 0, but not twice differentiable.

Things work differently in the complex plane.

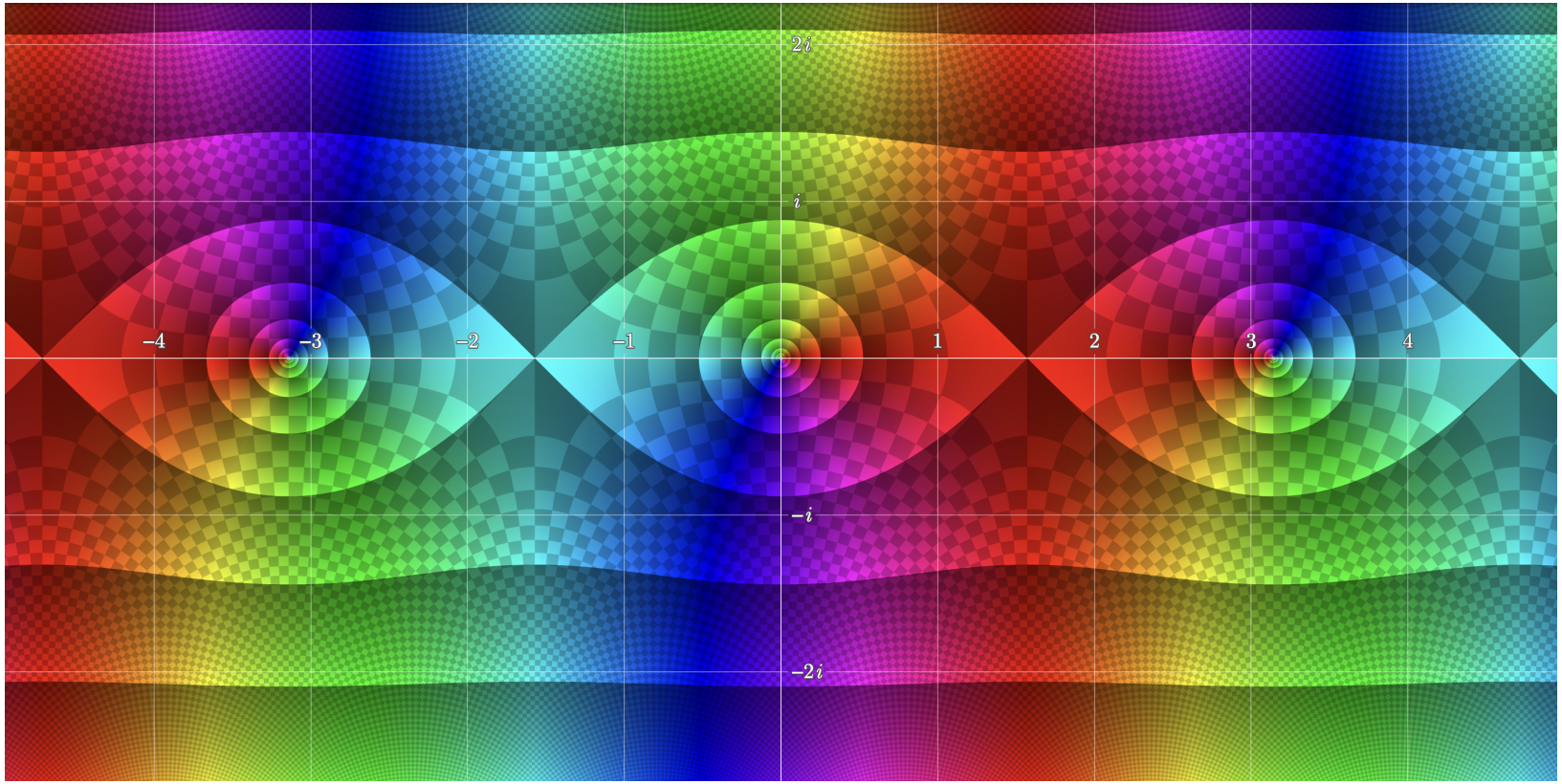
If a function is differentiable, it is infinitely differentiable.

In fact, it is analytic!



$\sin(x)$ is smooth and bounded in $\mathbb{R}$, but...

$\sin(x)$ is *not* bounded in the complex plane.



Why not?

Definition (Holomorphic function)

A function is *holomorphic* on a set $\Omega \subseteq \mathbb{C}$ if it is complex-differentiable at every point $z_0 \in \Omega$. That is,

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \text{ converges.}$$

Definition (Entire function)

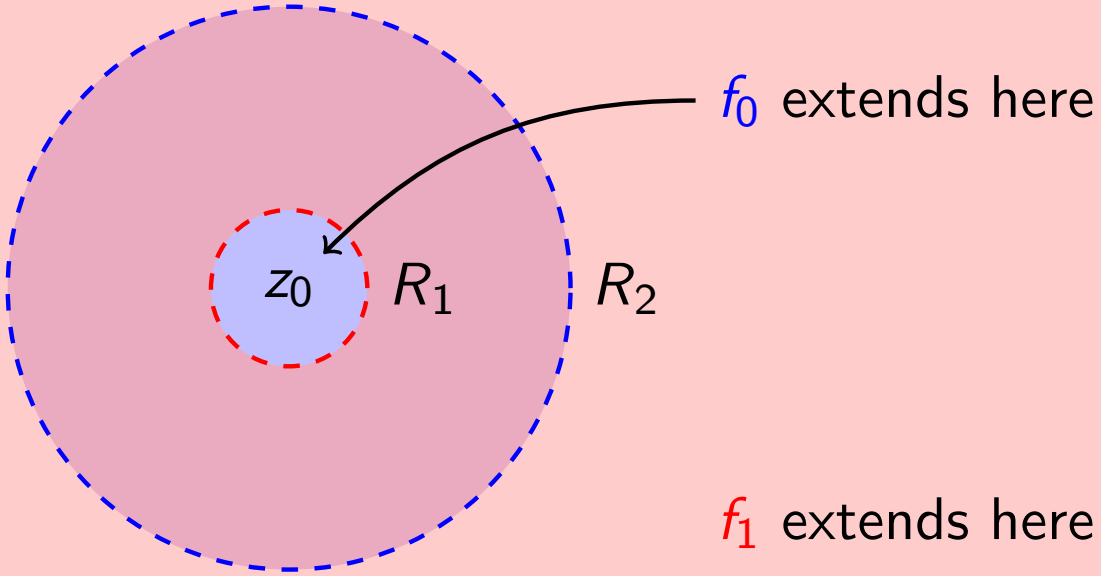
An *entire* function is holomorphic on the whole complex plane $\mathbb{C}$.

Theorem (Liouville)

Let $f : \mathbb{C} \to \mathbb{C}$ be entire and bounded. Then f is constant.

Theorem (Laurent decomposition)

Let $0 \leq R_1 < R_2 \leq \infty$. If f is holomorphic on the annulus $R_1 < |z - z_0| < R_2$ centred at z_0, then f can be decomposed as $f(z) = f_0(z) + f_1(z)$, where f_0 is holomorphic on $0 \leq |z - z_0| < R_2$ and f_1 is holomorphic on $R_1 < |z - z_0| \leq \infty$.



How do we integrate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{z^2+1}\, dz$?

$$\frac{1}{z^2+1} = \frac{1}{(z-i)(z+i)} = \frac{i/2}{z+i} - \frac{i/2}{z-i}$$

The points i and $-i$ are isolated singularities.

Definition (Types of isolated singularities)

Suppose z_0 is an isolated singularity of a function f. Let

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

be the Laurent series around z_0 in a neighbourhood of z_0.

- z_0 is a *removable singularity* if $a_n = 0$ for all $n < 0$
- z_0 is a *pole* of order N if $a_n = 0$ for all $n < -N$ and $a_{-N} \neq 0$
- z_0 is an *essential singularity* if $a_{-n} \neq 0$ for infinitely many n.

In our case, i and $-i$ are poles of order 1, also called *simple poles*.

$$\frac{1}{z^2+1} = \frac{i/2}{z+i} - \frac{i/2}{z-i}$$

To compute $\displaystyle\int_{-\infty}^{\infty} \frac{1}{z^2+1}\,dx$, we find the *residue* of i and $-i$.

Definition

Suppose z_0 is an isolated singularity of f. The *residue* of f at z_0 is defined by $\mathrm{Res}_{z_0} f = a_{-1}$, where a_{-1} is the -1st term of the Laurent series of f around z_0.

Lemma

If z_0 is a simple pole of f, then

$$Res_{z_0} f = \lim_{z \to z_0} (z - z_0) f(z).$$

For i, the residue is

$$\mathrm{Res}_i f = \lim_{z \to i} (z - i) \frac{1}{(z-i)(z+i)} = \frac{1}{2i}.$$

For $-i$, the residue is

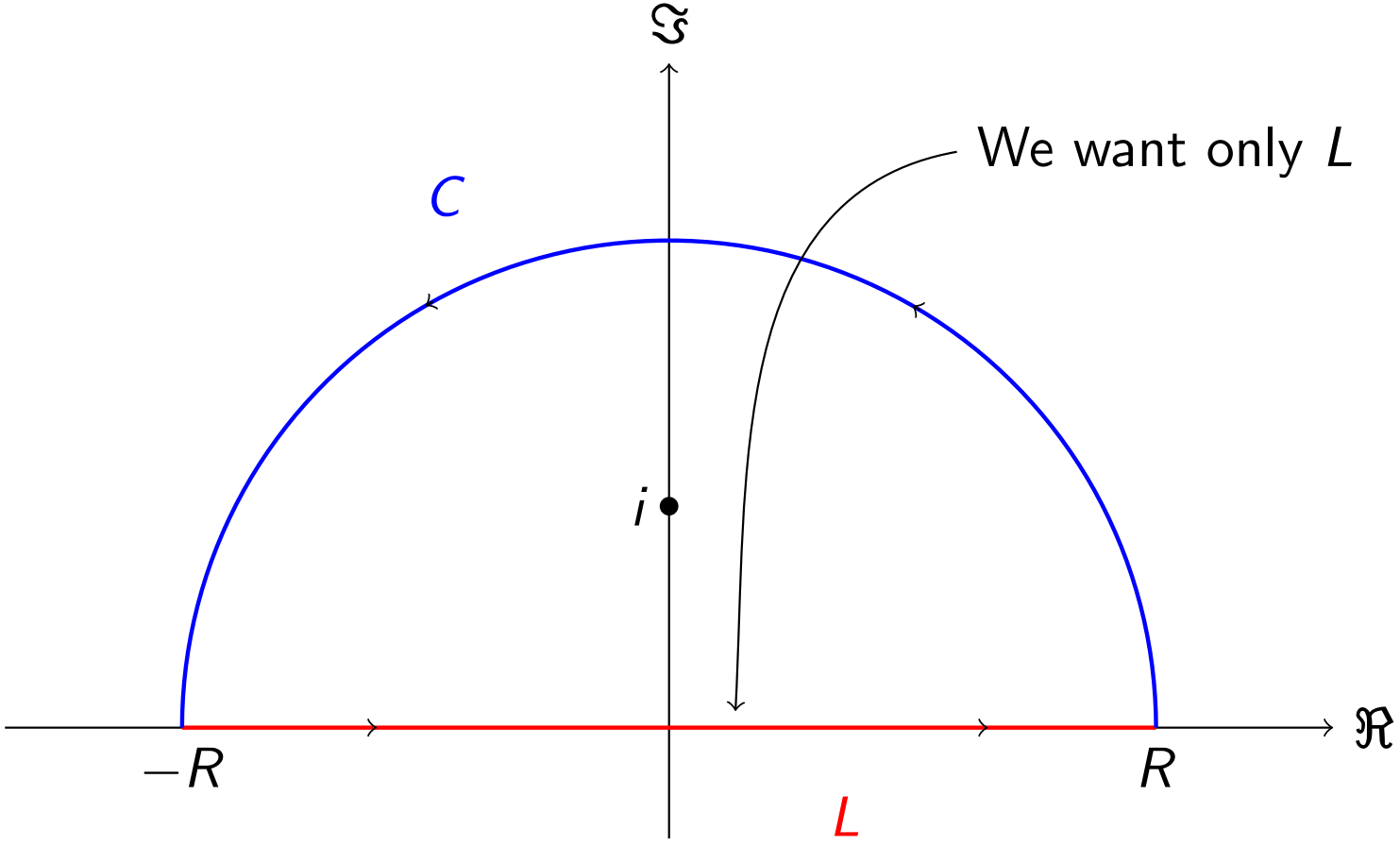
$$\mathrm{Res}_{-i} f = \lim_{z \to -i} (z + i) \frac{1}{(z-i)(z+i)} = -\frac{1}{2i}.$$

How does this help us?

Corollary (of the Residue Theorem)

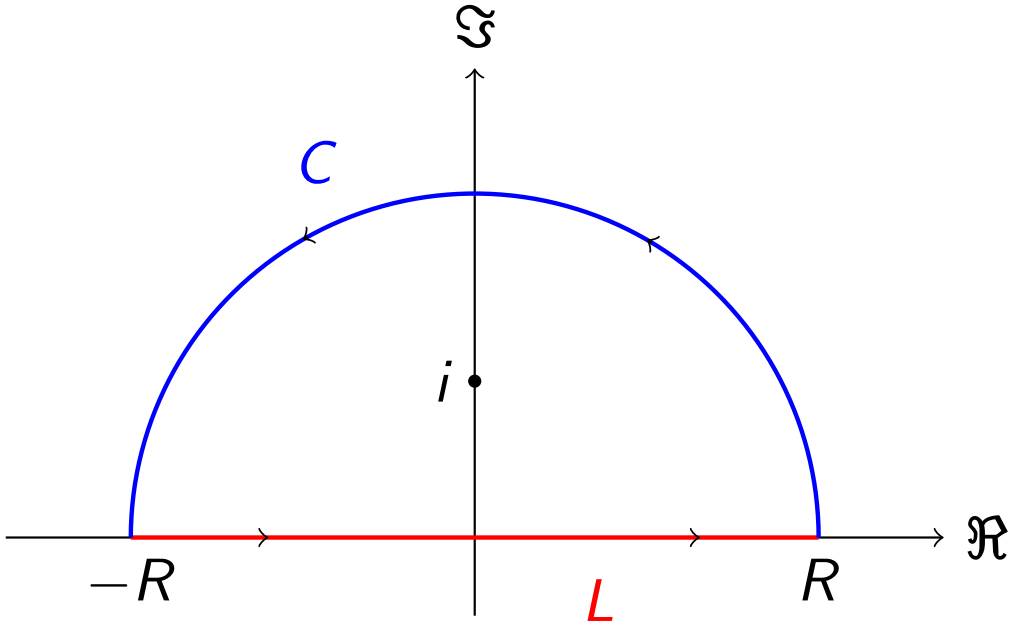
Let γ be a continuous closed curve with interior Ω. Suppose f is holomorphic on $\gamma \cup \Omega$ except for at isolated singularities $a_1, \ldots, a_n \in \Omega$. Then

$$\oint_\gamma f(z)\, dz = 2\pi i \sum_{k=1}^{n} Res_{a_k} f.$$



$$\int_{-\infty}^{\infty} \frac{1}{z^2+1}\, dz = L = \lim_{R\to\infty} \int_{-R}^{R} \frac{1}{z^2+1}\, dz$$

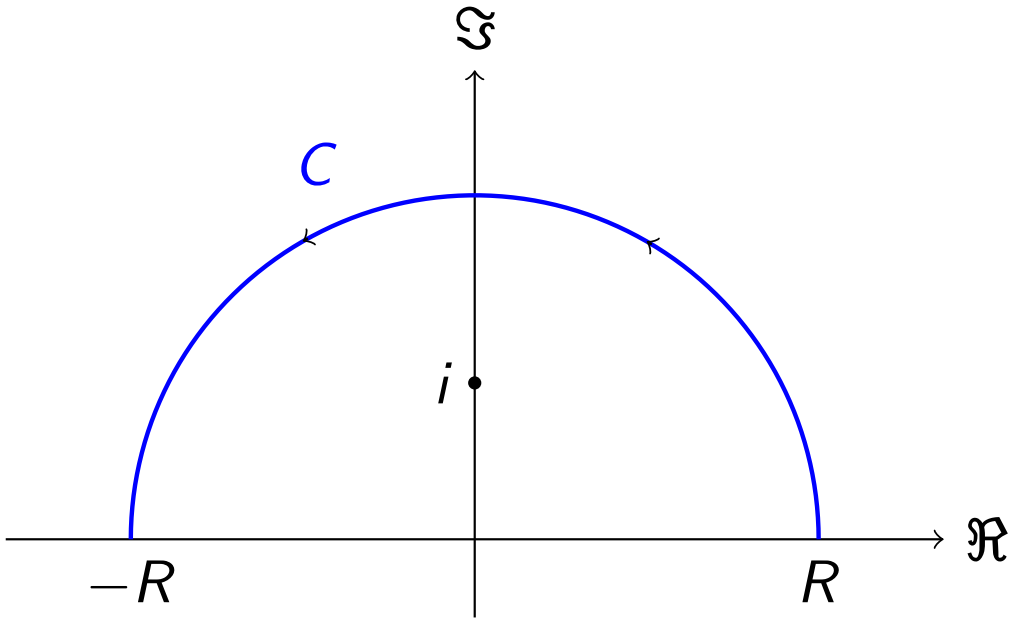
Our curve only contains the pole i, so only $\mathrm{Res}_i f$ contributes.



$$\int_{C \cup L} \frac{1}{z^2+1}\, dz \;=\; 2\pi i\, \mathrm{Res}_i f \;=\; 2\pi i \frac{1}{2i} \;=\; \pi.$$

What happens on C and on L?

Consider only the semicircular curve C.



We may parameterise C using $z = Re^{it}$ for $t \in [0, \pi]$, yielding

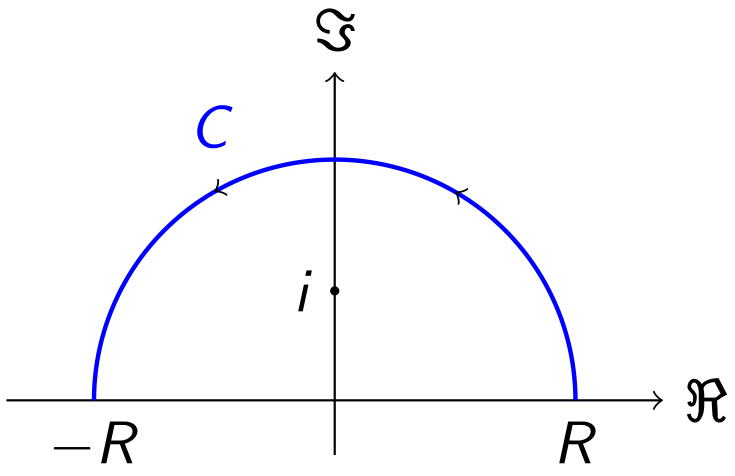
$$\int_C \frac{1}{z^2+1}\, dz = \int_0^\pi \frac{iRe^{it}}{R^2e^{2it}+1}\, dt.$$

Lemma (ML bound)

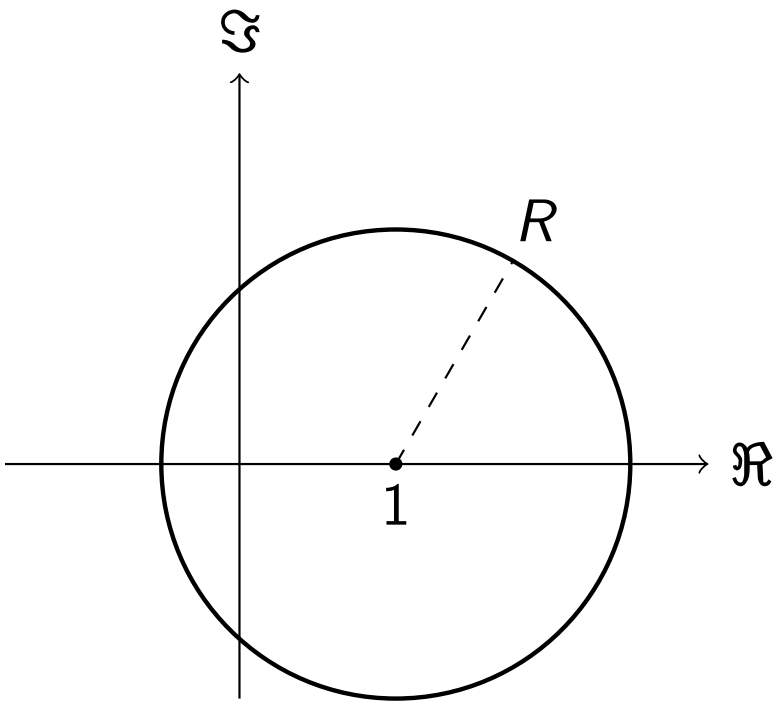
$$\left|\int_\gamma f(z)\,dz\right| \le \int_\gamma |f(z)||dz| \le ML,$$

where $M = \max_{z\in\gamma} |f(z)|$ *and* L *is the length of the path* γ.

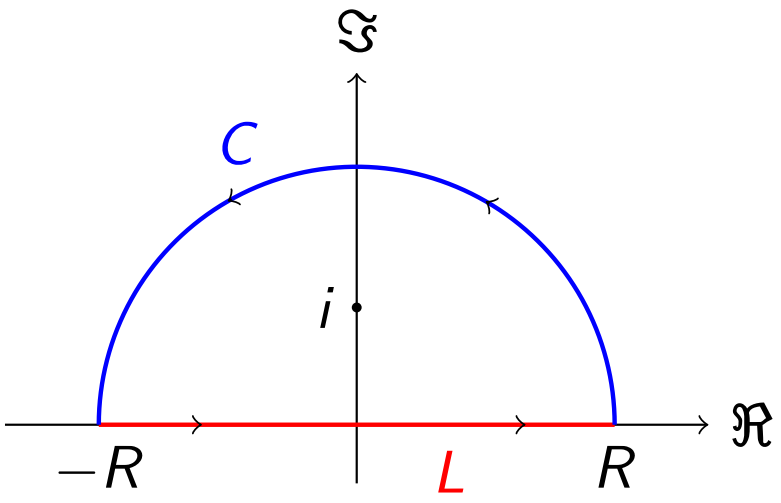
Since C is a semicircle, $L = \pi R$. What is M?



$$\left|\int_0^{\pi} \frac{iRe^{it}}{R^2e^{2it}+1}\,dt\right| \le \int_0^{\pi} \frac{R}{R^2-1}\,dt = \frac{\pi R}{R^2-1} = \frac{L}{R^2-1}.$$



So $M = \frac{1}{R^2-1}$, and our ML bound is $\frac{\pi R}{R^2-1}$.



As $R \to \infty$, $ML \to 0$, meaning that

$$\lim_{R\to\infty} \int_C \frac{1}{z^2+1}\, dz \;\leq\; \lim_{R\to\infty} \frac{\pi R}{R^2-1} \;=\; 0.$$

C contributes nothing!

$$\pi = \int_{C\cup L} \frac{1}{z^2+1}\, dz = \int_C \frac{1}{z^2+1}\, dz + \int_L \frac{1}{z^2+1}\, dz = 0 + \pi.$$

I want to thank my mentor and friend, Ilir Ziba, for an incredibly rewarding semester.

His excitement, patience, and intuitive explanations carried me through this material and helped me develop as a math student.

Yippee!